PRIMA
PARTNERSHIP FOR RESEARCH AND INNOVATION IN THE MEDITERANEAN AREA

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CRS4 IDEAS BECOME LIFE

soils4med

SOIL health monitoring and information systems FOR sustainable soil management in the MEDiterranean region

MedSIS manual

Administrators

Roberto Demontis

DATASETS | MAPS | DOCUMENTS | GEOSTORIES | DASHBOARDS

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1. Overview

A Soil Information System (SIS) is an integrated information system designed to facilitate the storage, analysis, management and dissemination of soil data.

To assess and monitor soil health and promote sustainable soil management in the Mediterranean region, there is a pressing need to improve the availability and accessibility of soil data.

One of the objectives of the SOILS4MED project (SOIL health monitoring and information systems FOR sustainable soil management in the MEDiterranean region) which addresses this need by designing and developing a decentralised network of country-based soil information systems.

This network aims to improve the interoperability, management, governance, and utilization of soil data across the Mediterranean region by establishing a federated architecture of SISs, starting with the development of a harmonized data model aligned with the World Reference Base for Soil Resources (WRB 2022), alongside the implementation of core functionalities for the ingestion, integration, and management of both legacy and newly acquired soil data.

The SIS is essentially a **Web GIS Catalogue** of resources related to soil data. It is based on Geonode (https://geonode.org/), whose resources include datasets, maps, documents, geostories and dashboards. In addition to the standard catalogue functionalities for managing **soil data**, MED-SIS includes an additional tool, known as the **Back Office**

MED-SIS staff use the Back Office to create and configure catalogue resources and to manage them through the Web GIS catalogue.

The **Med SIS** was designed and implemented by **CRS** (https://www.crs4.it/) (Roberto Demontis, Eva Lorrai, Laura Muscas, Piergiorgio Palla) with the support of all project partners.

The code is located in a GitHub repository: https://github.com/crs4/med_sis licensed under the GNU General Public License v3.0 (GPL-3.0) see LICENSE file for the full text.

Further details on Web GIS Catalogue functionalities can be found on the official GeoNode documentation page at https://docs.geonode.org/en/latest.

Video and manuals are accessible in the “Help” page of the MED-SIS or in the Github repository.

Soils4Med is a project funded by ***PRIMA*** (https://prima-med.org/) - Partnership for Research and Innovation in the Mediterranean Area.

This manual has been created as a reference guide to assist Med-SIS administrators with system installation and management, and is divided into four main sections: "Installation," "Management," "Administration Panel," and "Data Backup and Recovery."

2. Installation

The main components of the Med-SIS are two:

Component	Role	Stack
Web GIS Catalogue	Soil data management	Geonode (https://geonode.org)
Back Office	Web utilities for Soil data management	Nextjs (https://nextjs.org)

The **Web GIS Catalogue** is a Geonode WEB GIS catalogue for Soil data management

Main libraries:

Geonode 5.1.0
Geoserver 2.28.4
Django 5.2.17
Django Rest Framework 3.17.2
Python 3.12

The **Back Office** is a NextJS application that use the PrimeReact REACT Diamond template

Main libraries:

React 18.3.0
NextJS 15.5.15
Primereact 10.2.1

Getting started

Med-SIS is a Docker Compose multi container application tested in a Ubuntu server 24.04
The minimal requirements are:

Operating system: Ubuntu server 24.04
Storage : 1 TB
CPU: Intel 64-bit, 4.0 GHZ, 8 cores, 2MB Cache
The **RAM**: 32 GB
Software: Docker Compose (Desktop or Engine)

Docker Compose is a tool for defining and running multi-container applications. It is the key to unlocking a streamlined and efficient development and deployment experience.

Compose simplifies the control of your entire application stack, making it easy to manage services, networks, and volumes in a single YAML configuration file. Then, with a single command, you create and start all the services from your configuration file.
The link https://docs.geonode.org/en/latest/setup/docker/prerequisites/ might be useful for installing Docker.

Step clone the repository

```bash
git clone https://github.com/crs4/med_sis.git
cd med_sis
```

Step copy the .env.sample file in a new file named .env

```bash
cp .env.sample .env
```

Step edit the environment variables in the new .env file

Use a system editor and consult the comments in the file.

Step run docker compose build

```bash
docker compose build
```

Step run the docker compose application

```bash
docker compose up -d
```

Step access and test the site

Using the browser and the configured address and domain

3. Management

The Admin Panel permits configuring the geonode catalogue. Only staff user can access to the panel

Admin Panel Access:

• URL: http://<host_name>/admin

• Menu: Via the "Admin" link in the top user menu.

• Permissions: Requires the "staff" or "superuser" flag in the user profile.

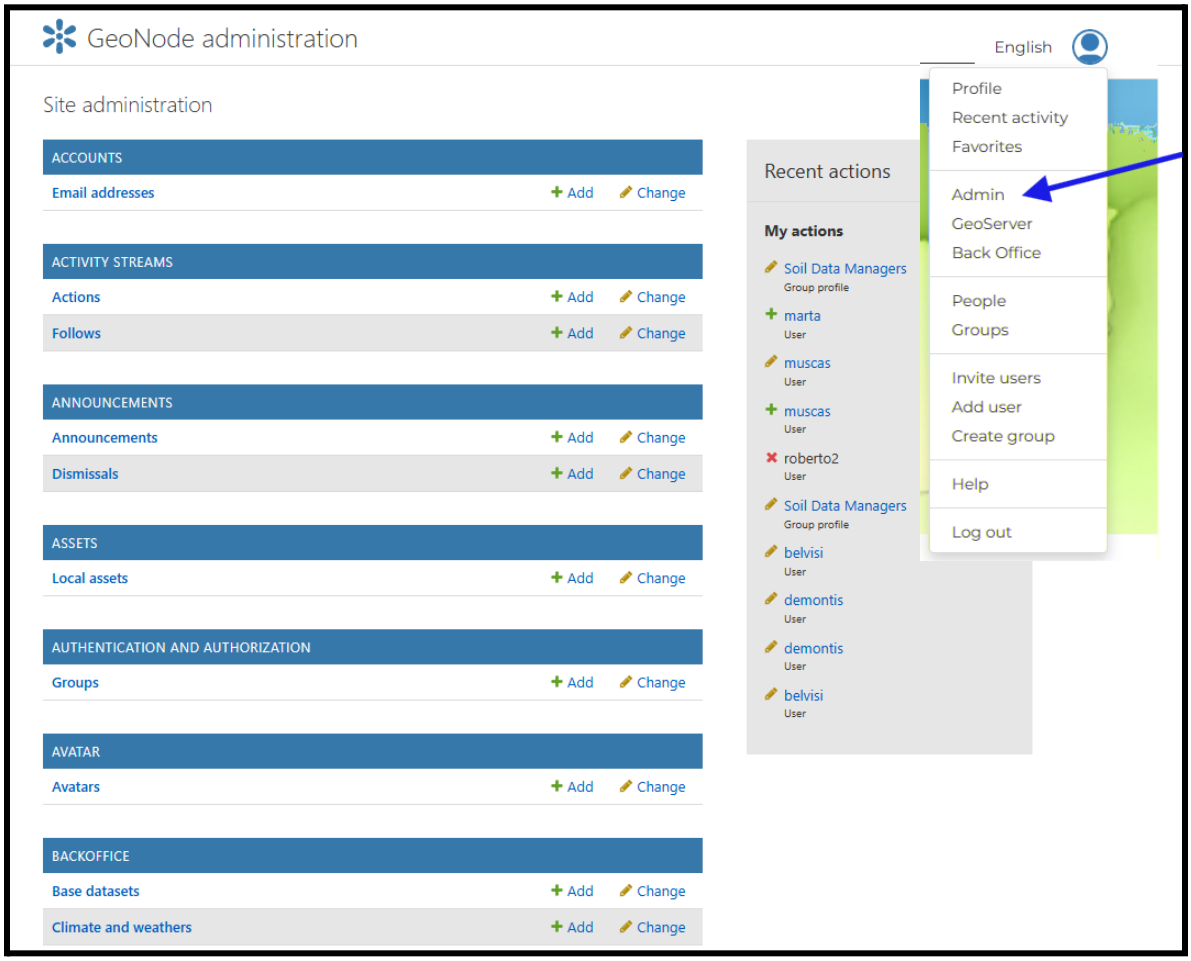


Figure 1 The Admin panel

see the GeoNode documentation at: https://docs.geonode.org/projects/v4/en/4.4.x/admin/index.html

USERS

The user can be managed using the Admin Panel or in the “People” page of the Catalogue.

Med-SIS staff have the necessary permissions to open the "People" or "Groups" pages (Figure 3) using the user menu at the top of the page.

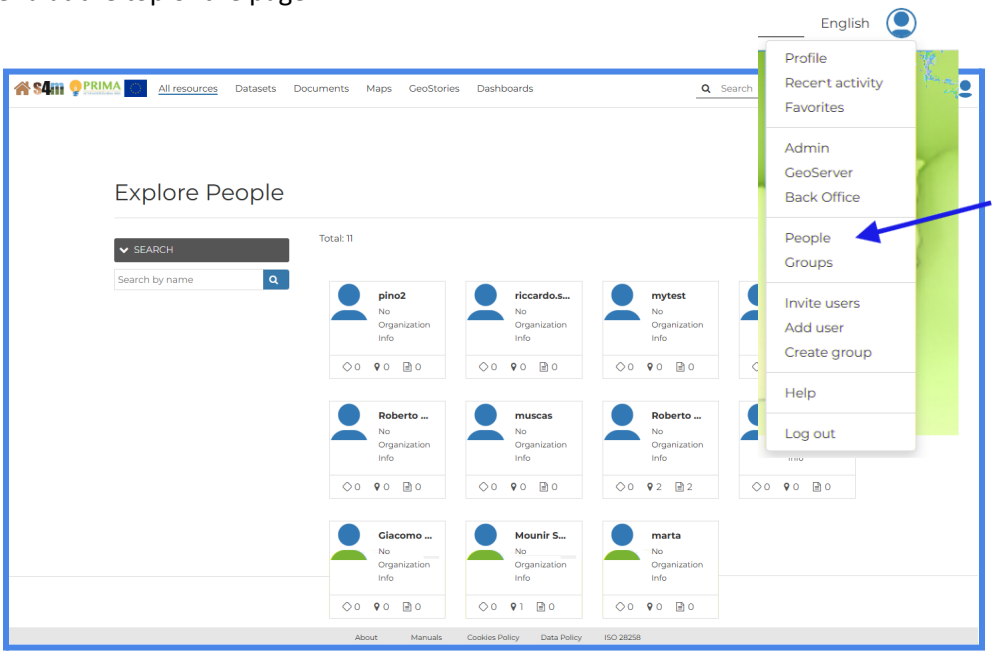


Figure 2 The People page

Clicking on a user card opens the corresponding profile.

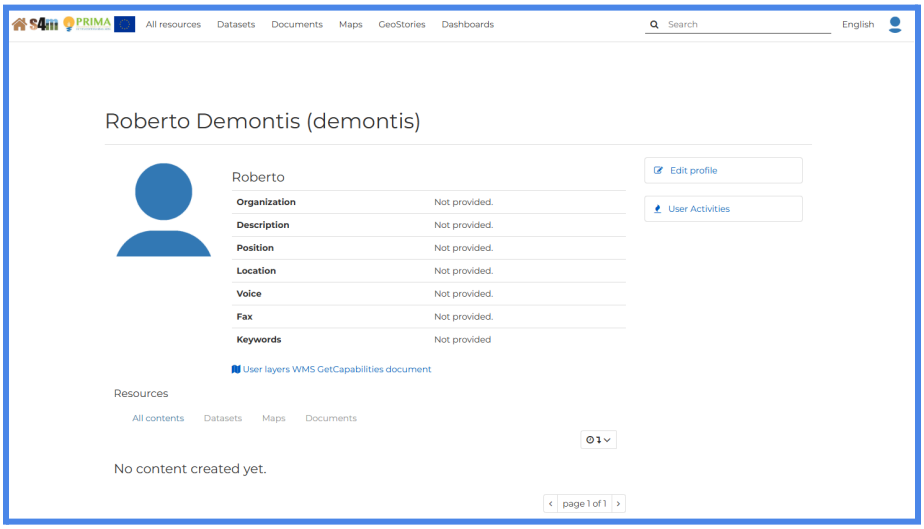


Figure 3 user profile

On the user profile page, you can edit the profile and view the user's activities and datasets.

The Admin panel permits to modify more other user properties.

Figure 4: Link to users list in the admin panel

Figure 5: List of users

The user list contains general information about registered users:

ID: the id of the user
USERNAME: username
ORGANIZATION NAME: organization name of the user
EMAIL ADDRESS: email address of the user
FIRST NAME: first name of the user
LAST NAME : last name of the user
STAFF STATUS : indicates whether or not the user is a staff member.
ACTIVE : indicates whether the user is enabled or not.

The Action drop down menu permits to delete the selected user or to set resource permission.
The button "Add User" permits to

The self registration is closed
(.env: REGISTRATION_OPEN=False)
The user can register in the system independently and is **active** after registration
(.env: ACCOUNT_APPROVAL_REQUIRED=False)
(.env: REGISTRATION_OPEN=True)
User registration must be approved by the MED-SIS staff and **isn't active** prior to approval
(.env: ACCOUNT_APPROVAL_REQUIRED=True)
(.env: REGISTRATION_OPEN=True)

A registered user can access all resources for which "View," "Download," "Edit," or "Manage" permissions have been set for the REGISTERED USERS group or for the user themselves or for a group of which the user is a member.

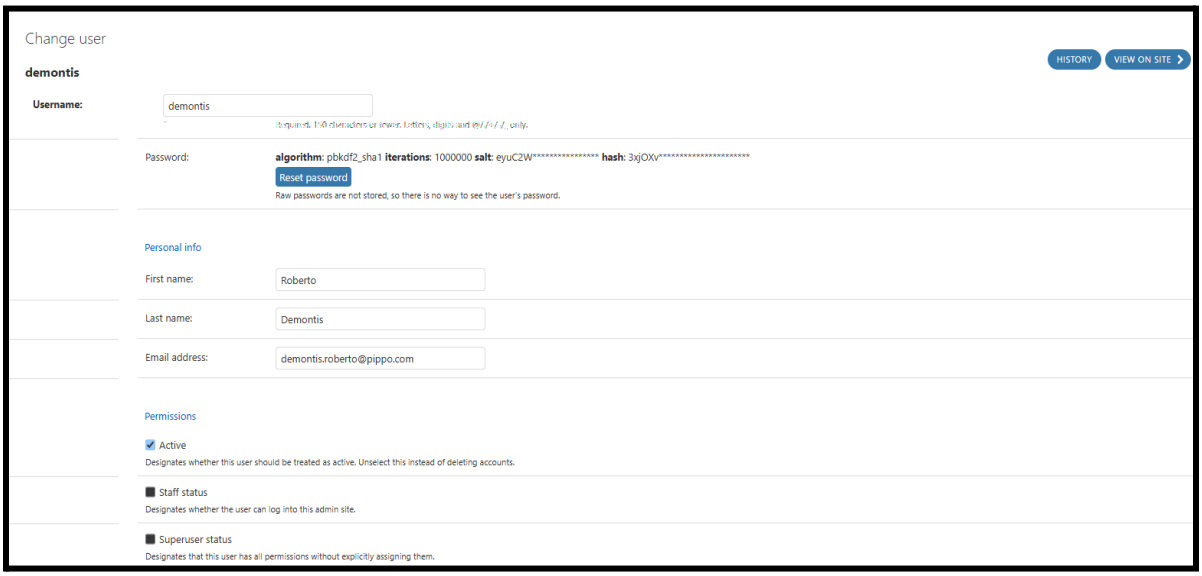


Figure 6: Change user - general info

The "Change User" shows a form that permits to reset password and manage the user status:

ACTIVE: the user is active and can access to the catalogue
STAFF STATUS: Designates whether the user can log into the admin panel
SUPERUSER STATUS: the user is an administrator with full permissions

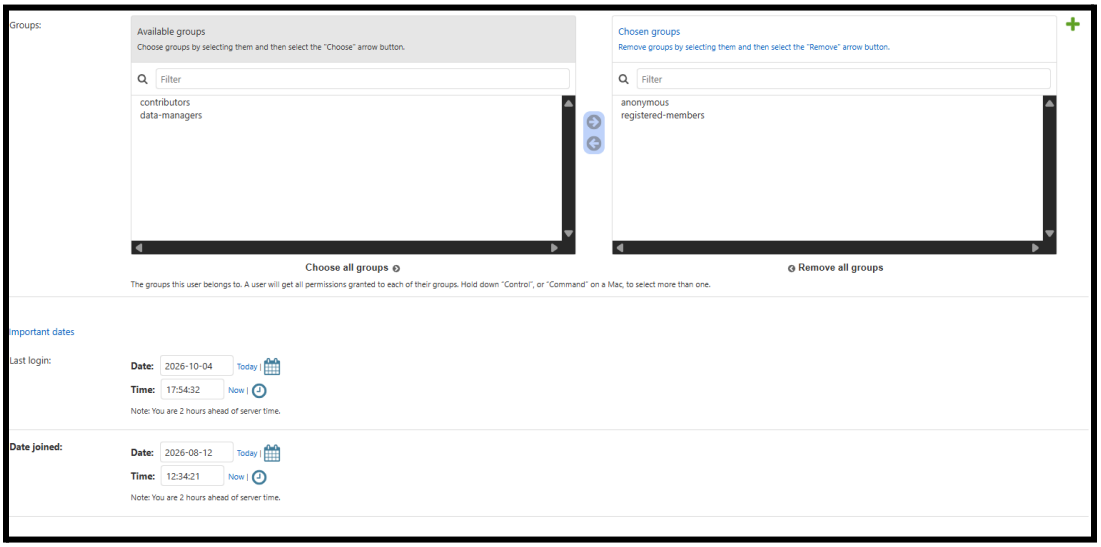


Figure 7: Change user - groups setting

Also is possible to assign the user to groups (Figure 7) and change profile (Figure 8). The panel in the bottom of the page permits to save changes or delete the user.

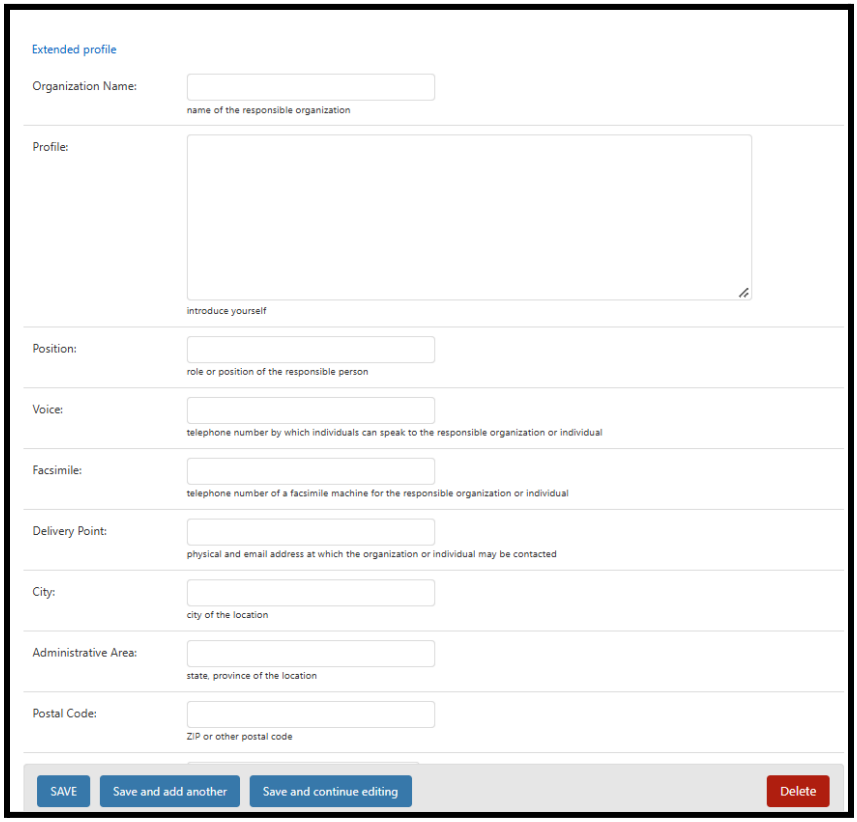


Figure 8: Change user - extended profile

GROUPS

The user can be managed using the Admin Panel or in the “Groups” page of the Catalogue.

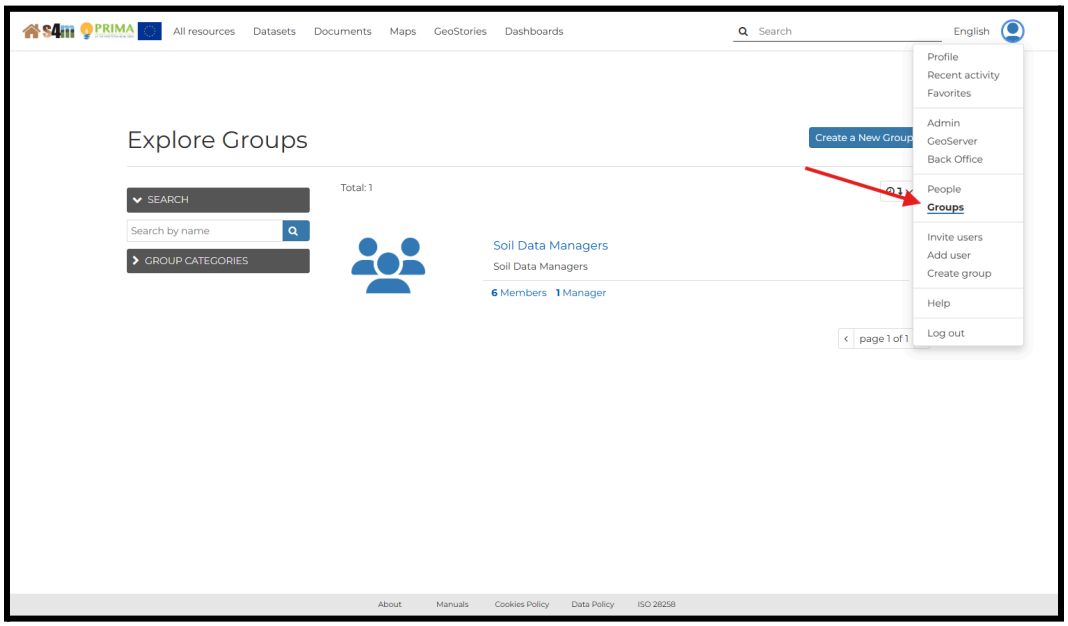


Figure 9: list of groups

Clicking on a groupcard opens the corresponding profile.

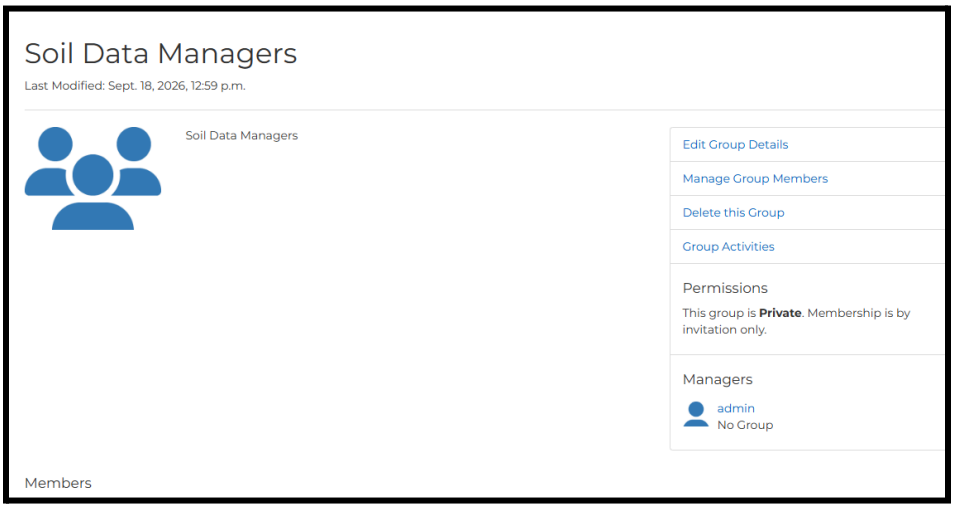


Figure 10: group description

The page in Figure 7 shows the members of the group and the managers. The panel on the right permits to perform some actions on group

Edit group details
Manage group members
Show group activities
Delete the group

Group **permissions** constitute the rule for member involvement. A group can be **private** or **public**.
Adding a member to a **private** group requires staff intervention.
All registered users can join a **public** group without moderation.

All group permissions are inherited by all users who are members of the group.

The Admin panel permits to modify more other group properties and permissions.

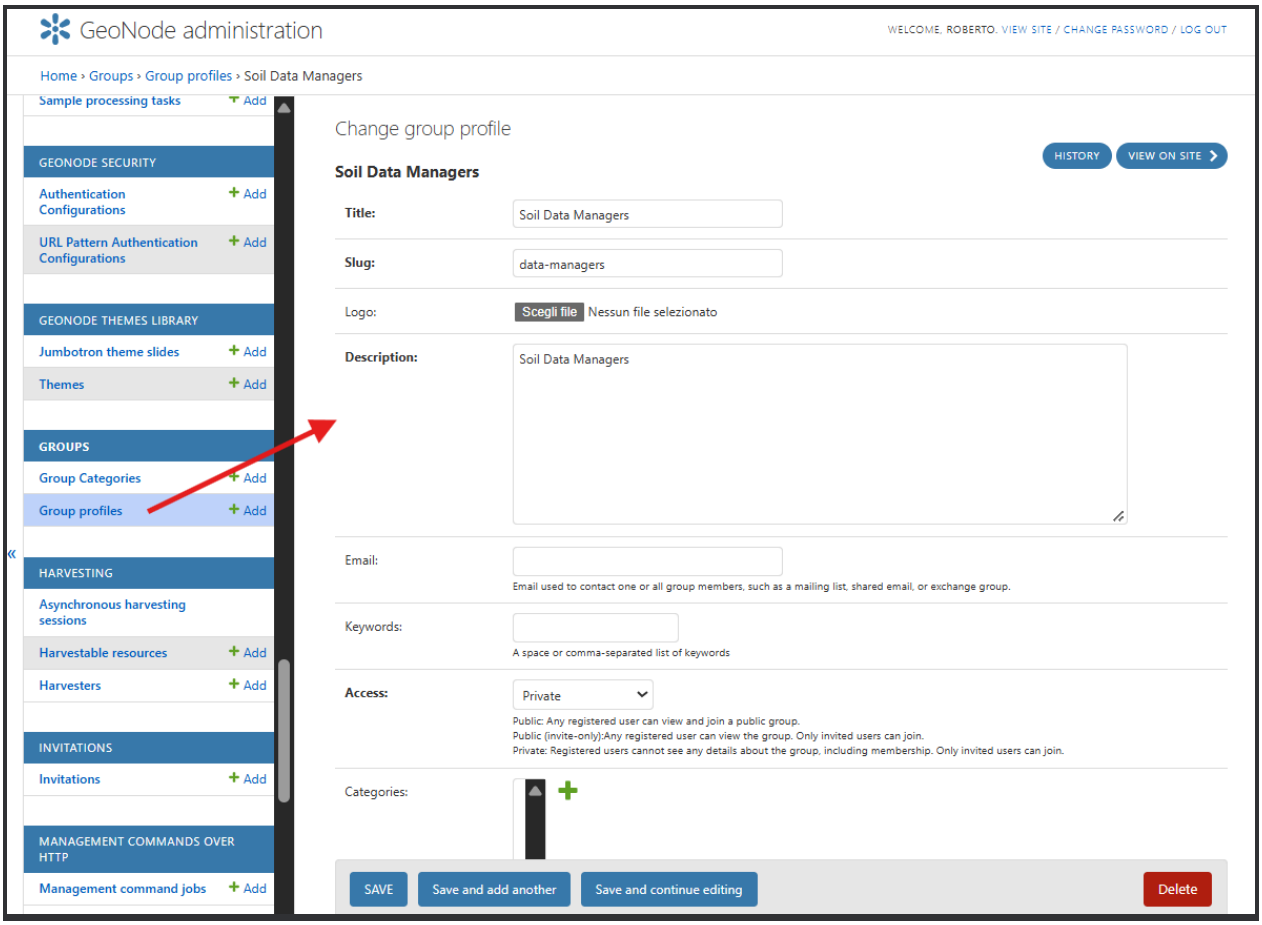


Figure 11: group profile

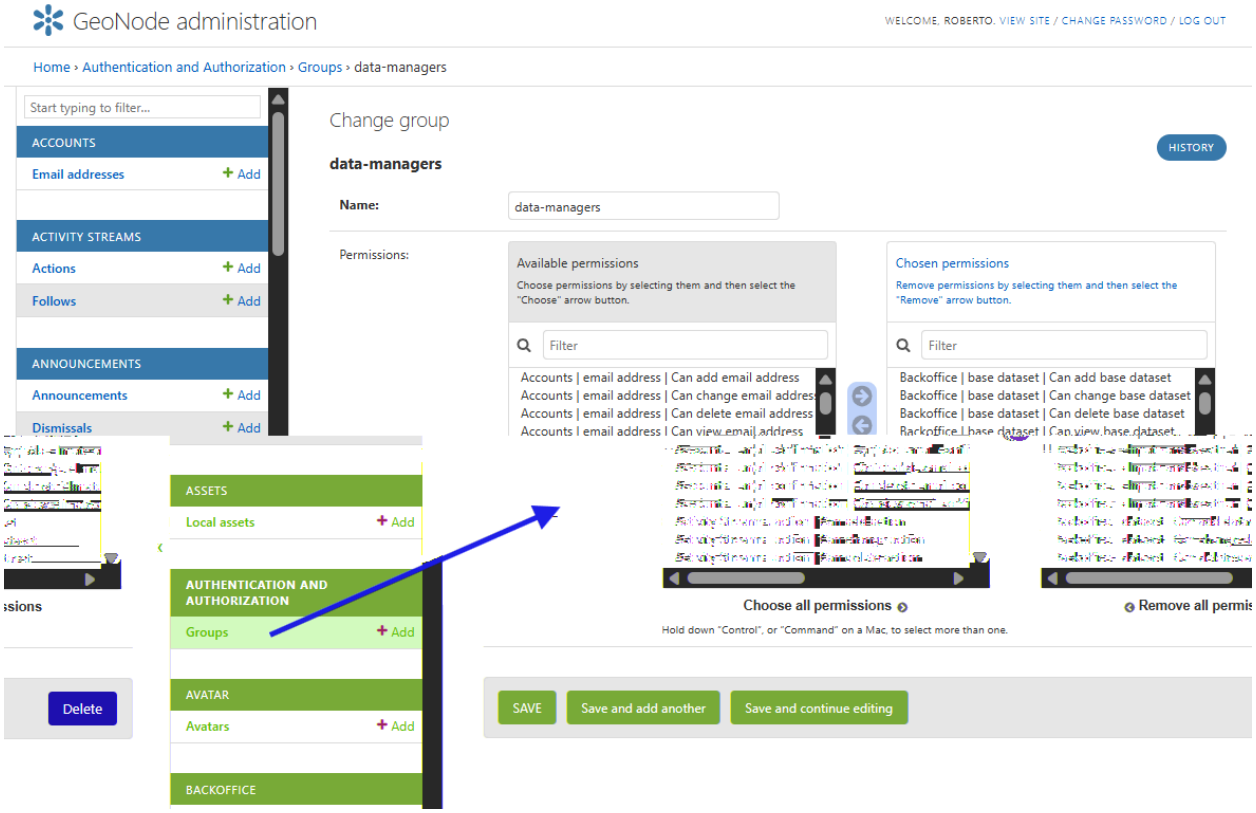


Figure 12: group permissions

SOIL DATA

Following installation, MED SIS requires the configuration and publication of the "**base datasets**", which enable the automatic updating of soil data whenever new data is uploaded.

*The "**base datasets**" should be accessible only to data managers and Med-SIS Staff*

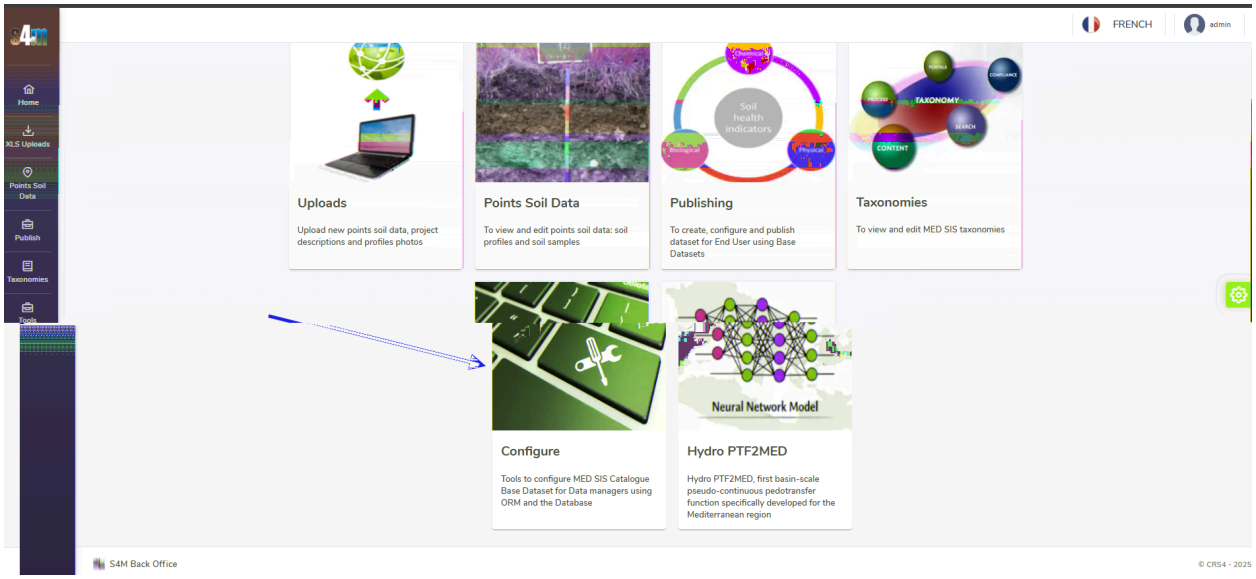
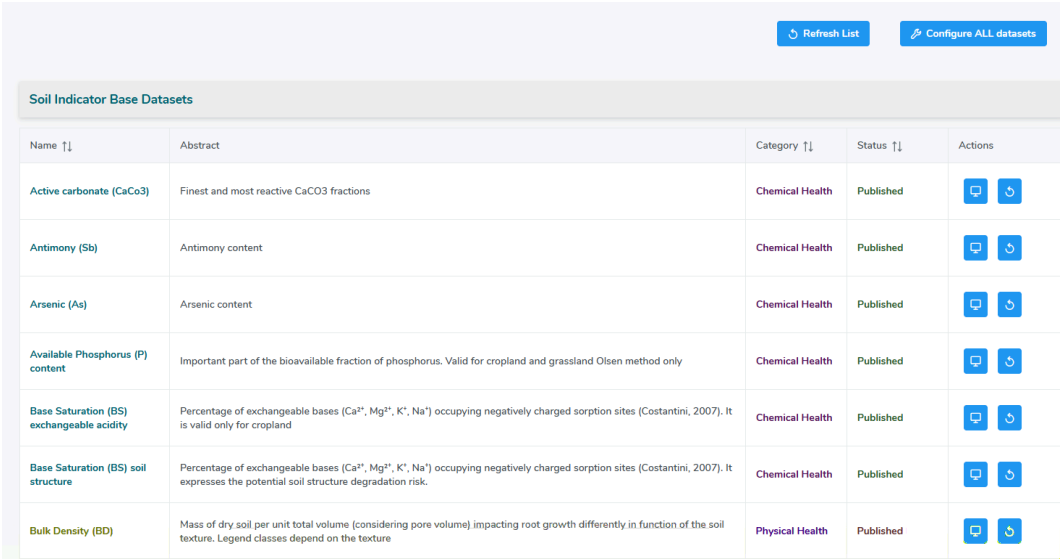


Figure 13: Back Office - Configure tool

The configure page shows two lists: **Soil Indicator base datasets** and Point **Soil Data Section Base Datasets.**

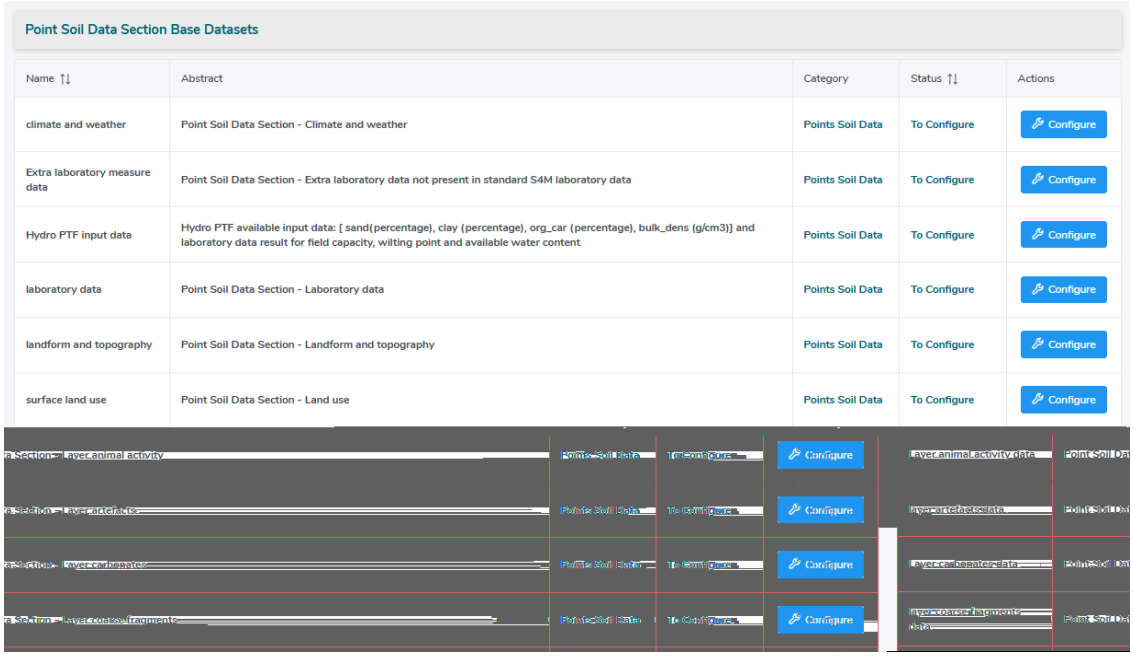


Refresh List | Configure ALL datasets

Soil Indicator Base Datasets

Name ↑↓	Abstract	Category ↑↓	Status ↑↓	Actions
Active carbonate (CaCo3)	Finest and most reactive CaCO3 fractions	Chemical Health	Published	
Antimony (Sb)	Antimony content	Chemical Health	Published	
Arsenic (As)	Arsenic content	Chemical Health	Published	
Available Phosphorus (P) content	Important part of the bioavailable fraction of phosphorus. Valid for cropland and grassland Olsen method only	Chemical Health	Published	
Base Saturation (BS) exchangeable acidity	Percentage of exchangeable bases (Ca^{2+}, Mg^{2+}, K^{+}, Na^{+}) occupying negatively charged sorption sites (Costantini, 2007). It is valid only for cropland	Chemical Health	Published	
Base Saturation (BS) soil structure	Percentage of exchangeable bases (Ca^{2+}, Mg^{2+}, K^{+}, Na^{+}) occupying negatively charged sorption sites (Costantini, 2007). It expresses the potential soil structure degradation risk.	Chemical Health	Published	
Bulk Density (BD)	Mass of dry soil per unit total volume (considering pore volume) impacting root growth differently in function of the soil texture. Legend classes depend on the texture	Physical Health	Published	

Figure 14: Soil Indicator base datasets



Point Soil Data Section Base Datasets

Name	Abstract	Category	Status	Actions
climate and weather	Point Soil Data Section - Climate and weather	Points Soil Data	To Configure	Configure
Extra laboratory measure data	Point Soil Data Section - Extra laboratory data not present in standard S4M laboratory data	Points Soil Data	To Configure	Configure
Hydro PTF input data	Hydro PTF available input data: [sand(percentage), clay (percentage), org_car (percentage), bulk_dens (g/cm3)] and laboratory data result for field capacity, wilting point and available water content	Points Soil Data	To Configure	Configure
laboratory data	Point Soil Data Section - Laboratory data	Points Soil Data	To Configure	Configure
landform and topography	Point Soil Data Section - Landform and topography	Points Soil Data	To Configure	Configure
surface land use	Point Soil Data Section - Land use	Points Soil Data	To Configure	Configure

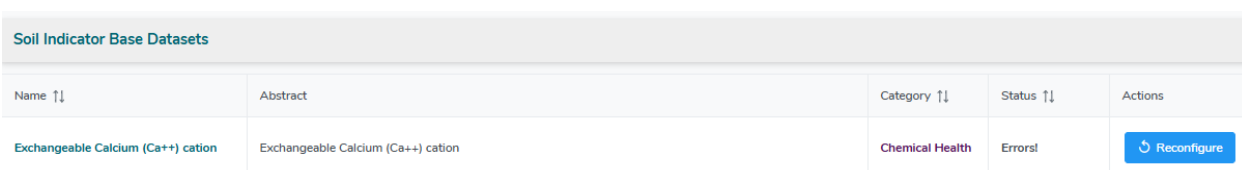
*Figure 15: Point Soil Data Section Base Datasets***.**

All the base datasets are by default loaded in the Geoserver but not in the Catalogue.

These datasets can be configured individually using the "Configure" button or collectively using "Configure all datasets" button.

Configuration **is required only once after installation**; it is performed on the server and may take several minutes and must therefore be carried out with caution.

Due to the number of datasets—approximately one hundred—using "Configure all datasets" may result in some errors. In this case, clicking the "Reconfigure" button can resolve the issue.



Soil Indicator Base Datasets

Name	Abstract	Category	Status	Actions
Exchangeable Calcium (Ca++) cation	Exchangeable Calcium (Ca++) cation	Chemical Health	Errors!	Reconfigure

Figure 16: The reconfigure button

The published base datasets are shown in the Catalogue pages; It is recommended to edit their metadata

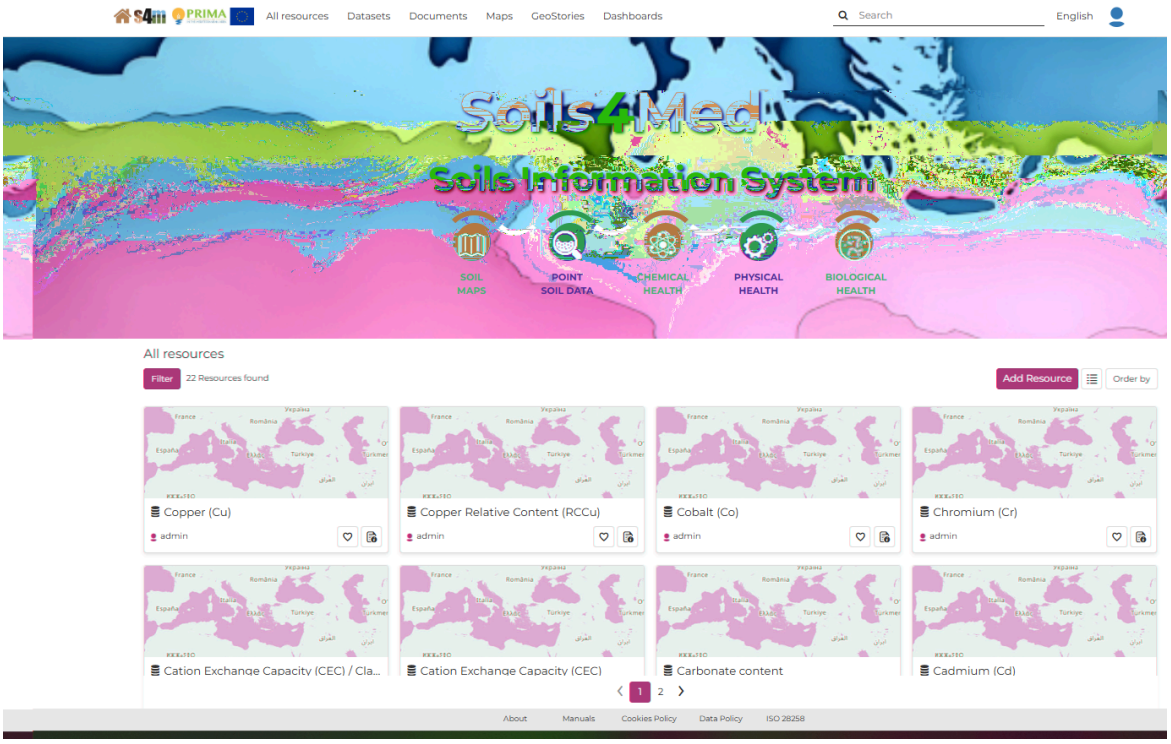


Figure 17: The list of base datasets

<u>The datasets may be empty because they are automatically generated based on the uploaded data.</u>

<u>It is recommended to use the catalogue pages for managing datasets</u>, but certain operations can be performed in the administration panel, as shown in Figure 14.

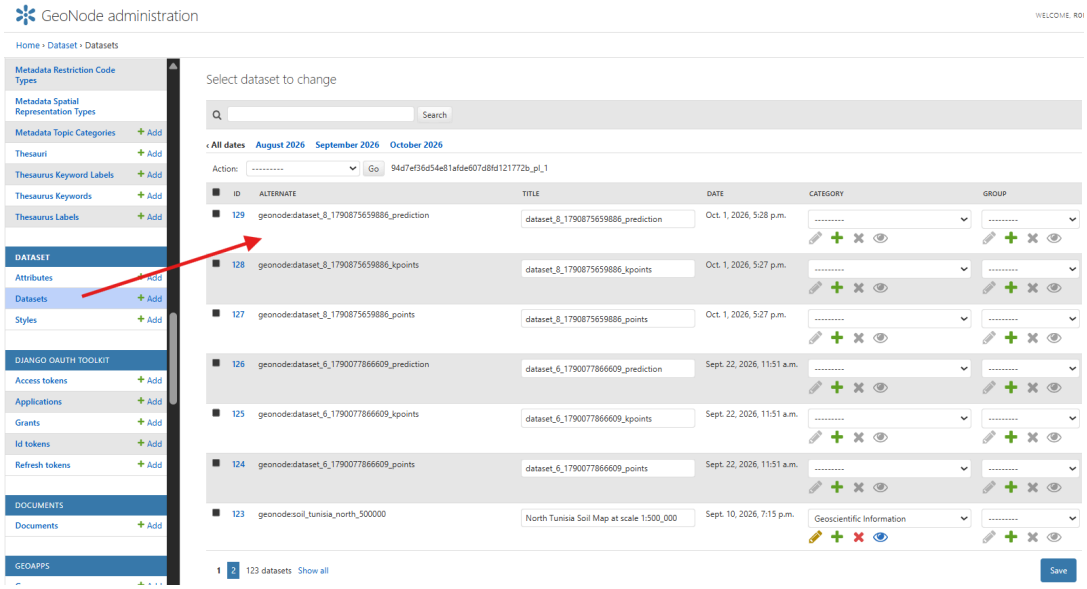


Figure 14: Admin Panel - Dataset Change

4. API

The MED-SIS uses API managing Catalogue resources and Soil Data

The Geonode API is described in several web document such as
https://docs.geonode.org/projects/v4/en/4.4.0/devel/api/V2/index.html

A dedicated API is available for managing soil data, accessible from the back office.:
/api/backoffice/

For security reasons, access to the APIs is restricted to the MED SIS Staff!

The data generated by the Soils Med project will be open and accessible via specialized APIs. These APIs, which comply with the ISO standard, are described in the document MED SIS MANUAL ISO .pdf.

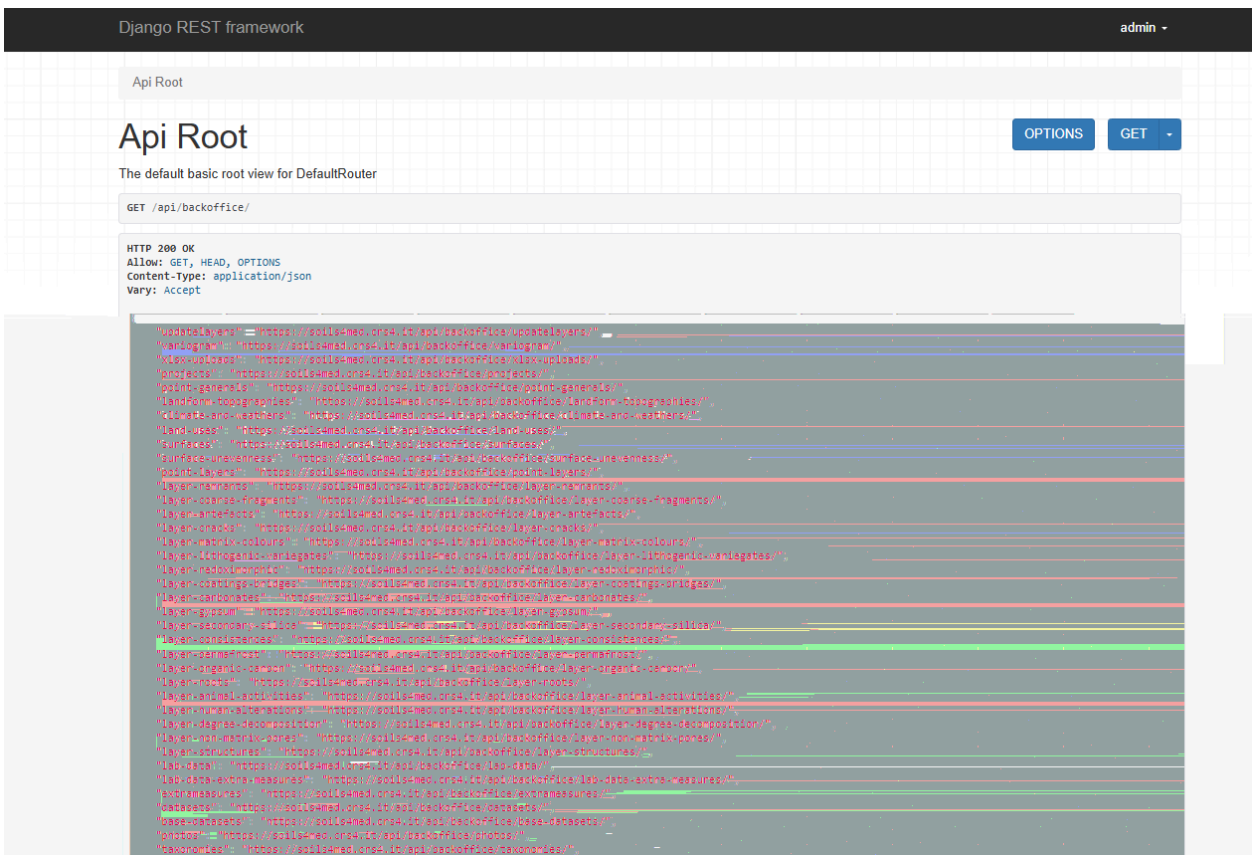


Figure 15: Back Office API

5. Backup and Restore

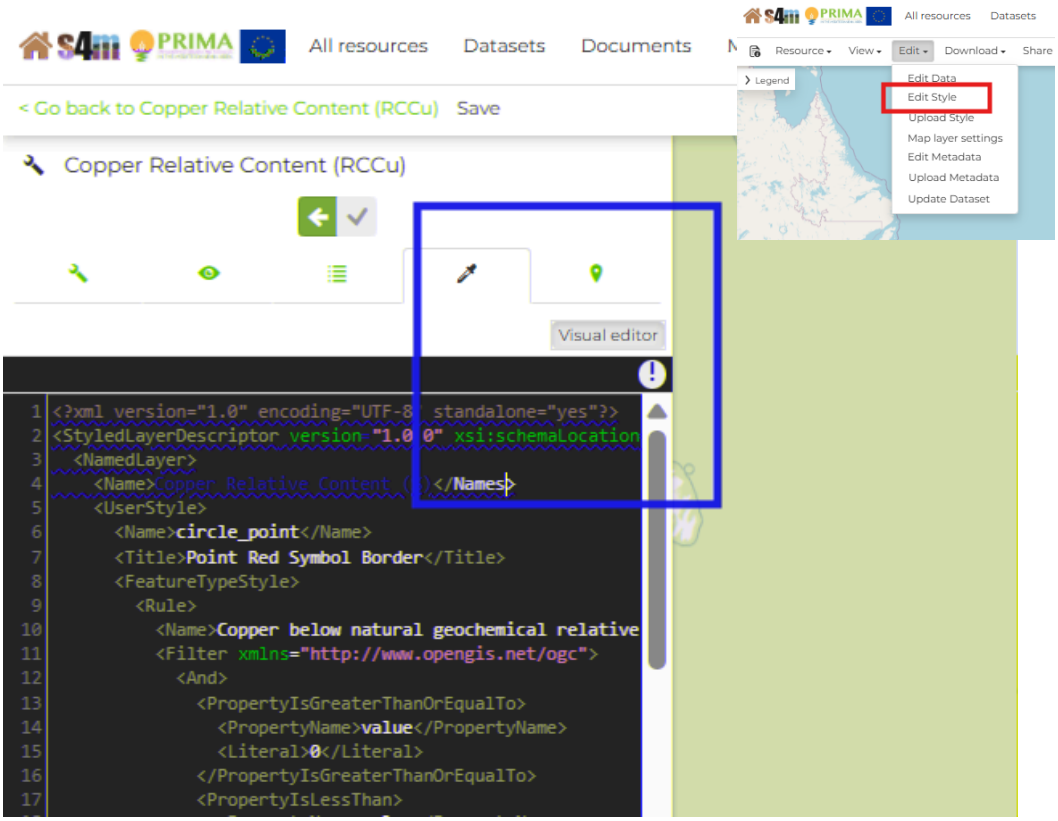
The backup and restore guide is available in the github repository at
https://github.com/crs4/med_sis/blob/main/BACKUP_RESTORE_GUIDE.md

6. Troubleshooting

- **The data and legend of a dataset is not displayed in views.**
  The legend's name shows a "!

  The problem may stem from:

  1) A problem with the dataset's style (SLD code)
     -> Go to the Style configuration and verify that is ok or not



  2) The communication between GeoNode and GeoServer regarding dataset permissions for groups.
     -> If the user is a "Data Manager," they can modify the dataset permissions by assigning themselves management permissions. If the user is not a "Data Manager," the layer permissions must be modified to grant the user "download" permission.

3) Outdated browser data when switching users;
   -> specifically, logging out as User A and then logging in as User B may result in User A remaining logged into GeoServer. To resolve the issue, open GeoServer and perform a logout.

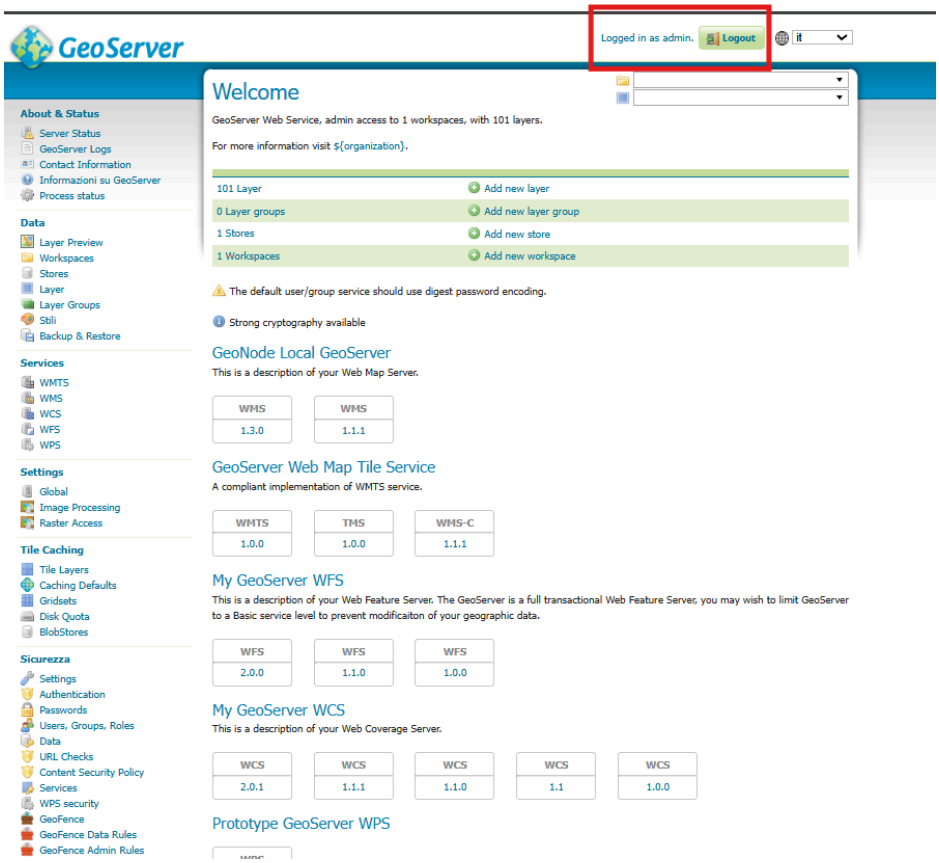


Figure 16: Geoserver log out